Measurement of radiation and particles

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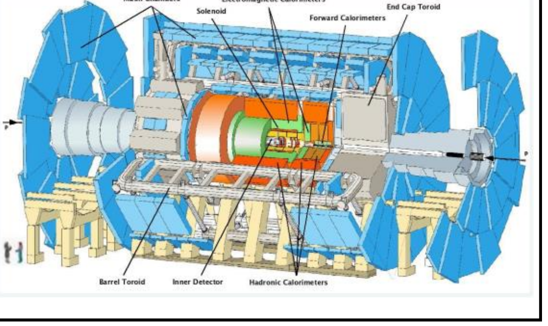
What is a particle / radiation detector?

Detectors can measure attributes of a particle such as
- energy, momentum, spin, charge
- particle type
- coordinates of interactions
- or: they simply register the presence of the particle

Most, but not all detector types create electronic signals. Examples of non-electronic detectors are
- photographic film
- cloud chamber
- bubble chamber
- thermoluminescent dosimeters

Particle detection (with electronic detectors) and measurement is based on
- interaction of particles with matter
- collection of created charges (or light)
- creating an electronic signal
- processing the signal
- visualizing the resulting data

The simplest and most complex particle detectors



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Particle detection: fundamental interactions of ionizing radiation and matter

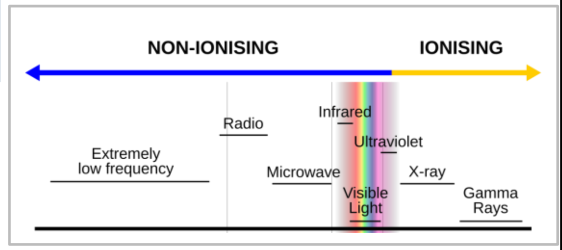
Ionizing radiation

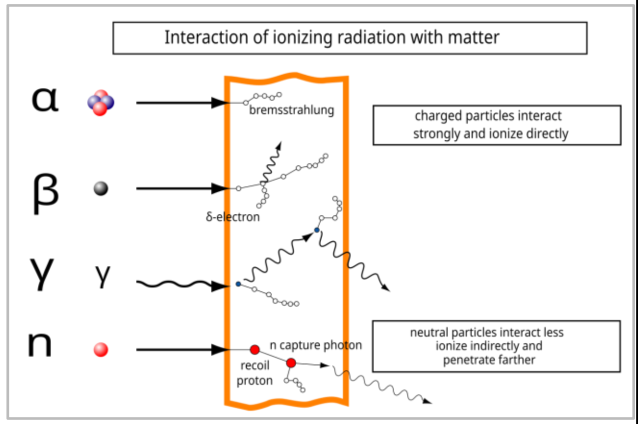
- radiation that causes the production of electron – ion pairs
- it consists of particles or electromagnetic radiation that have sufficient energy to ionize atoms

Particles

- alpha particles (^{4}He nuclei)
- beta particles (electrons or positrons)
- protons
- ions
- any other charged particles

These may occur as a result of radioactive decay.





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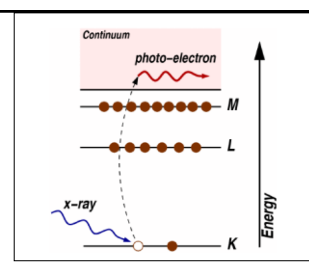
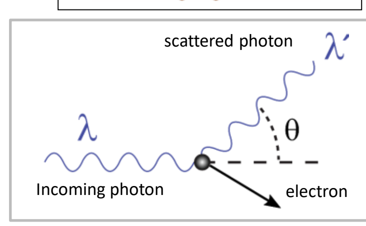
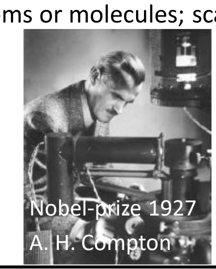
Particle detection: fundamental interactions of ionizing radiation and matter

Electromagnetic radiation (gamma-radiation or X-rays) may interact with matter in different ways:

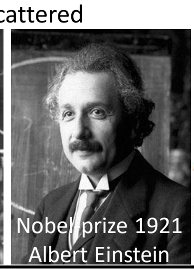
- Photoelectric effect
  - Emission of an electron from material caused by electromagnetic radiation; the electron is normally in a “close-to-nucleus” orbit; incident photon ceases to exist
- Compton-scattering
  - The photon interaction releases loosely bound electrons from the outer valence shells of atoms or molecules; scattered photon takes away some energy
- Pair production
  - An electron-positron pair is produced from a photon near a nucleus



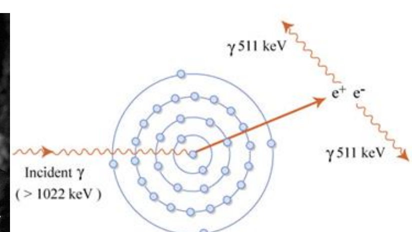




Nobel-prize 1927
A. H. Compton



Nobel prize 1921
Albert Einstein



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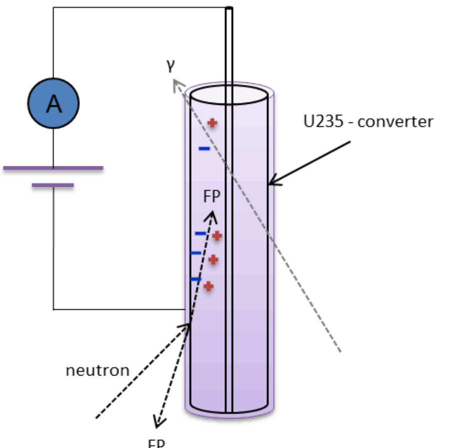
Particle detection: neutral particles (eg: neutrons)

- Neutrons have no charge so they are not able to ionize
- During detection we do not detect the neutron itself but a product of a reaction caused by the neuron
- Neutrons cause nuclear reactions (eg: fission)
- The product of these nuclear reactions is capable of producing very high level of ionization
- In this lecture, neutron detection is not discussed

Fission Chamber



Fission products cause a significantly larger amount of ionization within the fission chamber than the gamma radiation.

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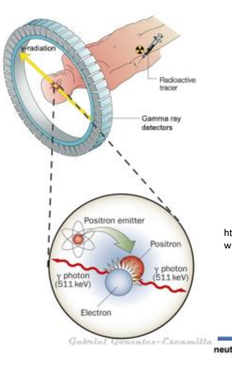
Where are radiation / particle detectors used?

Particle or radiation detectors are used *in a very wide range of applications*

- Medicine (mainly electromagnetic radiation)
  - CT, PET, SPECT scanners
  - dosimetry
- Nuclear applications (neutrons and photons)
  - Reactors I&C instruments (information, control)
  - Fusion research
  - dosimetry
- Fundamental physics research (all known particles)
  - CERN, ILL, ESS, accelerators, neutrons sources etc.
- Space research (high energy particles)
- Geophysics and geology research, measurements (neutrons, photons)
- Environmental measurements
- …



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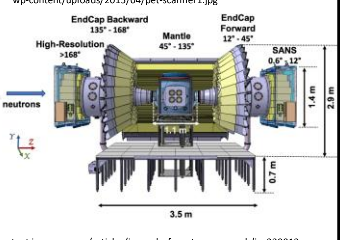
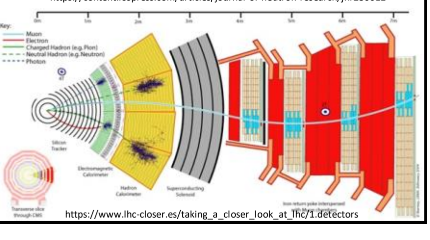


https://content.iospress.com/articles/journal-of-neutron-research/jnr230012



https://www.lhc-closer.es/taking_a_closer_look_at_lhc/1.detectors

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General characteristics of radiation detectors

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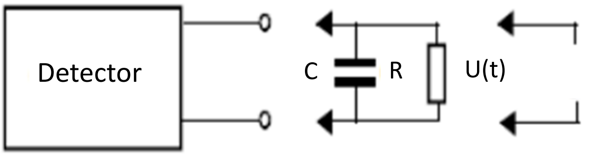
Characteristics of particle detectors

- They can operate in *pulse mode* or in *current mode*
- In pulse mode the measured pulses are used to create a spectrum
- They are characterized by:
  - dead-time, which is a temporary lack of ability to detect; it depends on the count rate of particle detection
  - energy resolution, which is determined mostly by the detector type
  - efficiency of detection, which is function of detector type, material, dimensions and measuring geometry
- Pile-up and escape phenomena are also important – not discussed further in this presentation

Simplified electronic models of detector + electronics:



- The amplifier has a certain equivalent capacitance *C* and resistance *R*. The capacitance is influenced by the cables, electrodes of the detector, input capacitance of amplifier etc.
- The detector also has some capacitance (eg. a gas-filled detector is in fact a capacitor).
- These characteristics determine how the produced electronic signal behaves.

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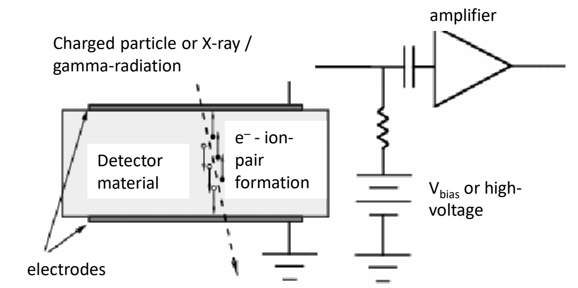
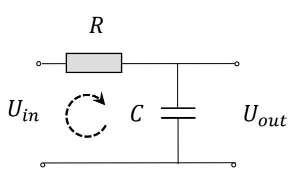
How is the electrical signal generated?



- Charged particle or ionizing electromagnetic radiation ➪ electronic charges are created in the detector material (these are called *primary charge carriers*)
- *Amount of charge* of primary charge carriers *is linearly proportional to incident particle energy* (not always fully true, but usually preferable)
- Collection of charges (required time may vary from millisecond for ionization chamber … to nanosecond in organic scintillators, semiconductor detectors)
- Current pulse ➪ voltage pulse on detector output ➪ pulse shaping and amplification ➪ shape of pulse depends on RC of equivalent circuit

Effect of RC on output voltage of detector



$$U_{in} = IR + \frac{Q}{C} \quad \Rightarrow \quad I = \frac{U_{in}}{R} - \frac{Q}{RC}$$

$$U_{out} = \frac{Q}{C} \quad \Rightarrow \quad U_{out} = \frac{1}{C}\int_0^T I dt$$

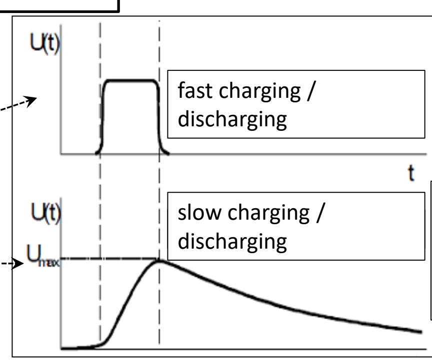
$$\boxed{\text{if} \quad \tau = RC >> 1} \quad \Rightarrow \quad \frac{Q}{RC} << \frac{U_{be}}{R} \quad \Rightarrow \quad \boxed{I \approx \frac{U_{be}}{R}}$$

$$U_{out} \approx \frac{1}{RC}\int_0^T U_{in} dt \qquad Q = \int_0^{t_c} I dt$$

$RC = \tau << t_c$

Charge collection time t_c is important

$t_c << RC = \tau$

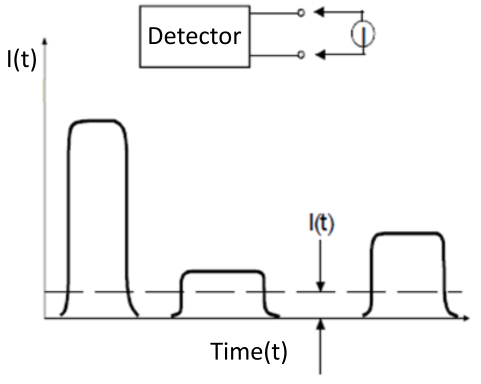


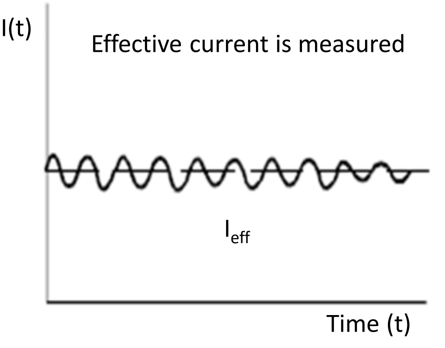
small *RC*: output voltage signal closely follows current of collected charge; preferred if timing is important or count rates are very high

large *RC*: output voltage has a maximum determined by total charge collected; *precise energy measurement is possible*

Current (integral) mode of operation

- Current mode: measurement of detector current averaged over a certain length of time ⇨ application in dosimetry or in cases when pulses are impossible to measure due to very high count rate
- Dosimetry: measurement of energy transferred by radiation to 1 kg of tissue





$$I_{eff}^2 = \sqrt{\frac{1}{T}\int_0^T I^2(t)\mathrm{d}t}$$

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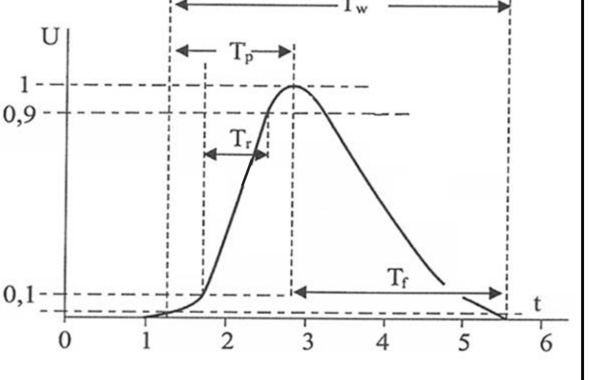
Detector operated in pulse mode

- Rise time = increase time of electronic pulse
- Decay time = decay time of the pulse
- Every single detection event is measured individually and the signal is processed (amplified, shaped …)
- The maximum of the output voltage is measured and converted to digital
- Where is the maximum? Derivate!
- Temporal behavior of signal carries important information
- Due to individual processing it is possible to make statistics of the measured pulses
- Accumulated data will create the *pulse height spectrum*



- T_w total pulse length
- T_p peaking time
- T_r rise time
- T_f decay time
- T_s shaping time

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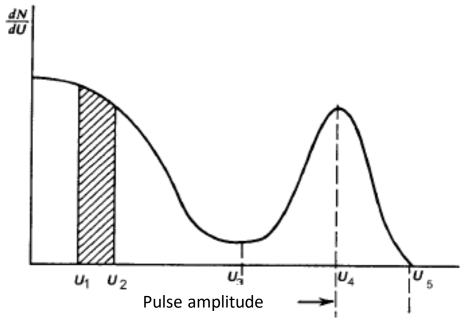
Pulse height spectrum: energy distribution of photons / charged particles detected

Definition of differential spectrum: $\Delta N = \int_{U_1}^{U_2} \frac{\mathrm{d}N}{\mathrm{d}U}\mathrm{d}U$



Horizontal axis: amplitude of detected pulses

Vertical axis: number of pulses detected for a given amplitude range

- Multichannel analyzer produces the spectrum
- Analyzer program is used to visualize the result

Why is there a peak?

- Many particles transfer the same amount of energy to detector ⇨ same signal amplitude is produced many times

Why is the peak broadened (not like a Dirac-delta)?

- The main reason is statistics. Every energy transfer process is a bit different, resulting in randomly varying amount of charge collected

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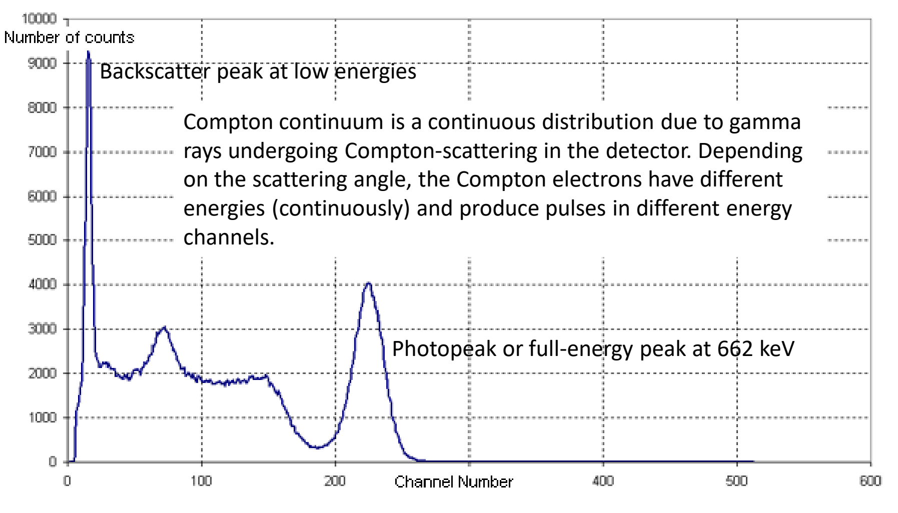
Structure of a pulse height spectrum



Spectrum of gamma photons detected by a scintillator detector. Photons are emitted by caesium isotope ^{137}Cs, which emits a single gamma line of 662 keV

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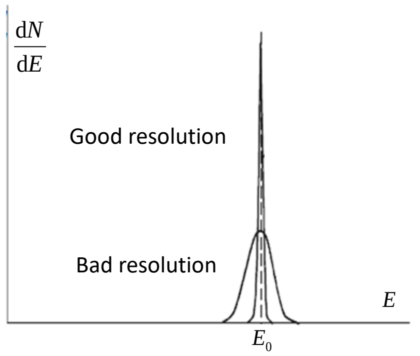
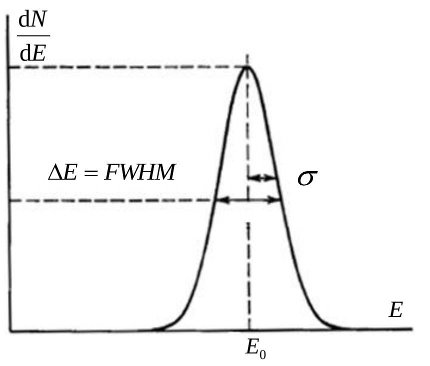
Energy resolution

$$R = \frac{\Delta E}{E_0} = \frac{FWHM}{E_0}$$

- Energy resolution is defined as the ration of width of peak to peak energy
- Peak shape follows Gaussian distribution $$I(E) = \frac{A}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(E-E_0)}{2\sigma^2}\right)$$





- Different detectors have different peak width.
  Worst: scintillation detector, medium: gas-filled, best: semiconductor detector

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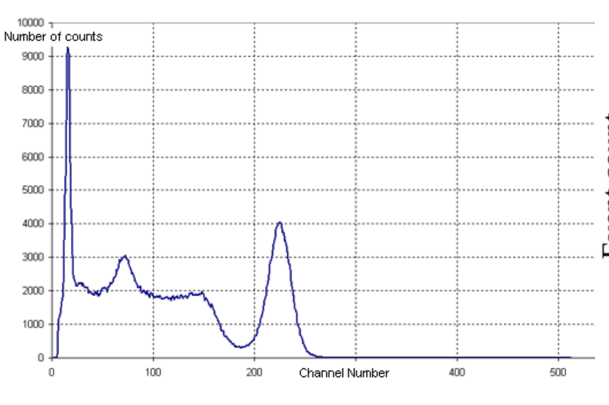
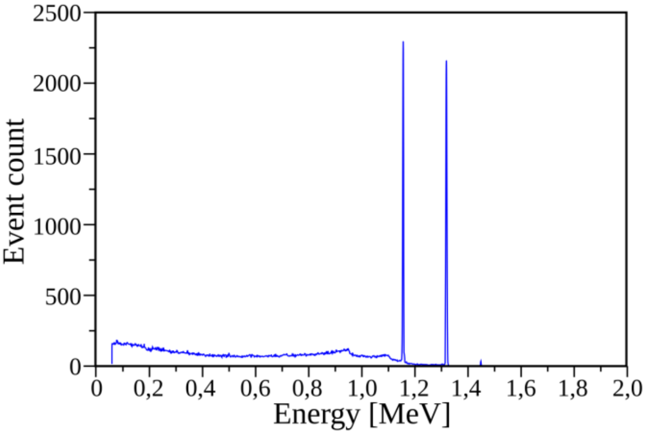
Comparison of spectra achieved by different detectors





Spectrum of gamma photons detected by a scintillator detector (left) and semiconductor detector (right). The difference of peak widths is significant. Semiconductor detectors are not discussed in detail in this lecture.

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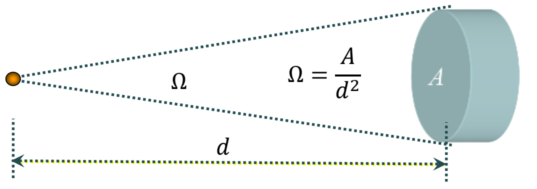
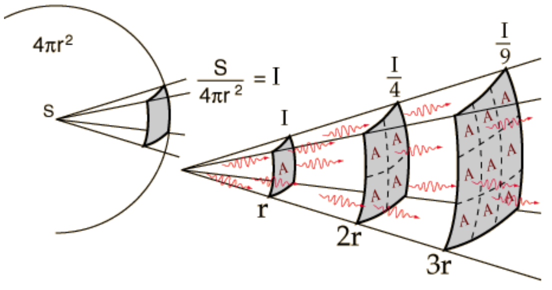
Detection efficiency

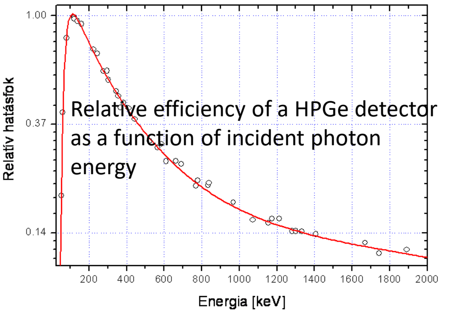
- Definition: $\eta = \dfrac{\text{Number of detected events}}{\text{Number of particles started}}$



- Depends on detection geometry: this determines how large portion of particles may encounter the detector material



- Detection events in the detector material: depends on likelihood of interaction of radiation and detector, which is material (Z, density) and energy dependent



Relative efficiency of a HPGe detector as a function of incident photon energy

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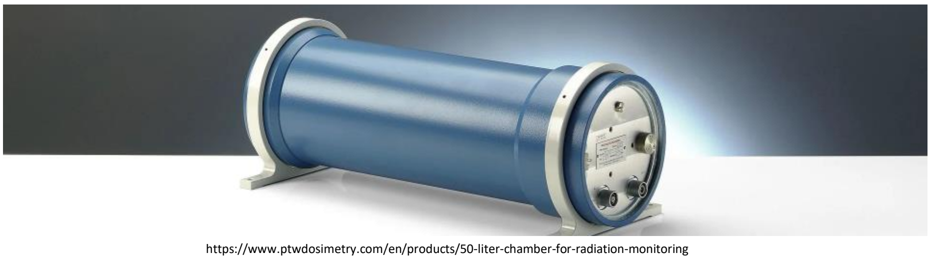
Specific detector types:

Gas-filled ionization chambers



https://www.ptwdosimetry.com/en/products/50-liter-chamber-for-radiation-monitoring

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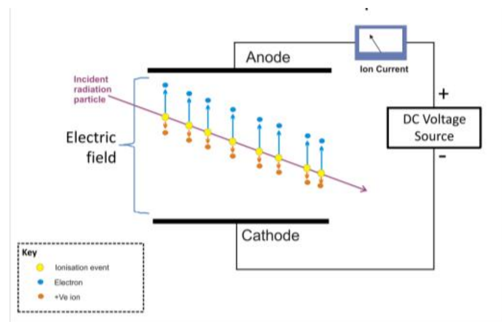
Gas-filled detectors

- Classification of gas-filled detectors
  - ionization chamber
  - proportional counter
  - Geiger-Müller (GM) tube



- First electronic sensor measuring ionization radiation
- Development: first half of the 20th century
- Simple structure and electronics

  ⇨ low cost
- Suitable to measure α, β, γ radiation

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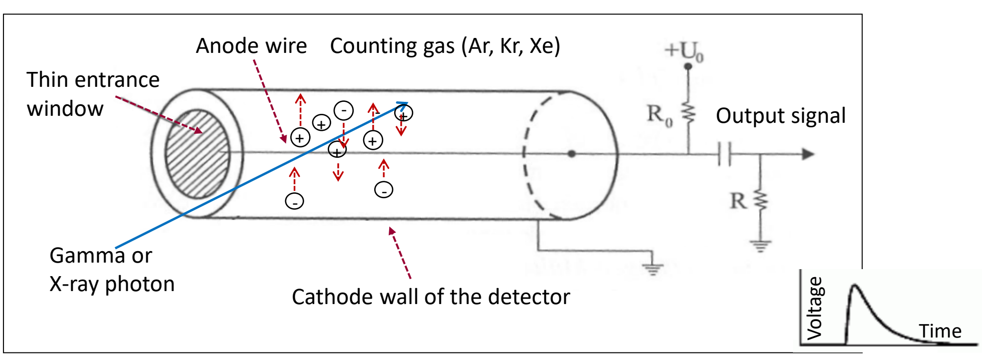
Structure of gas-filled detectors

- Ionization and excitation of gas atoms/molecules along the particle trajectory ⇨ positive ions and free electrons
- External electric field ⇨ (-) charges move to anode and (+) to cathode ⇨ production of electronic signal ⇨ $U_{anode}(t)$ ⇨ electric pulse
- Collected charge ~ ability to ionize ~ energy imparted by particle
- This means linearity ⇨ ability to measure particle energy



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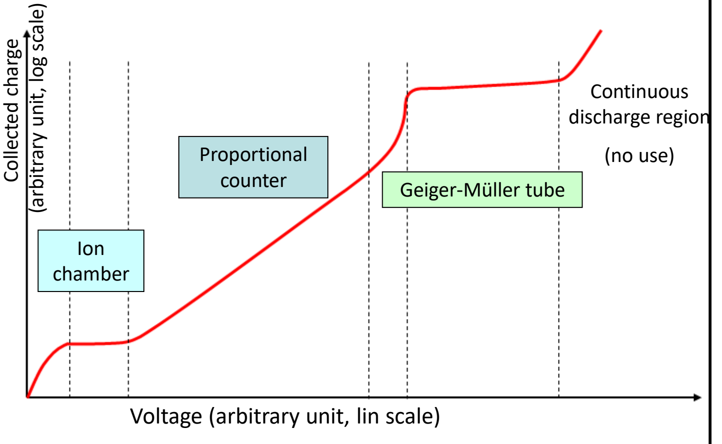
Variation of number of generated electron-ion pairs in gas ionization detectors

- The figure shows the characteristics if the applied voltage is changed
- In the *ionization chamber* range, all primary charges are collected as recombination is negligible
- *Proportional counter range*: there is significant secondary ionization due to high speed of accelerated electrons
- In Geiger-Müller range, extremely high number of secondary charges, creating an avalange

Assumptions:

- Geometry: cylinder form, anode wire
- Radiation: constant intensity



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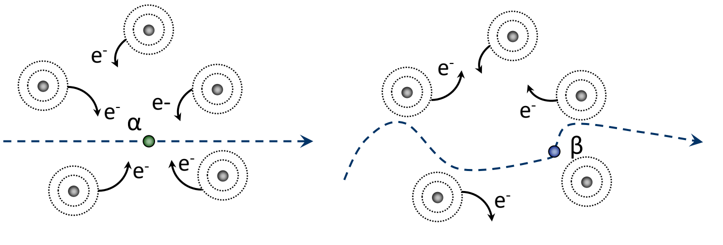
Gas ionization by charged particles and recombination processes

- Path of ionizing particle is
  - Straight line (almost) for heavy particles, such as alpha-particle or proton
  - Zig-zagged for electrons



- Equation of recombination: $\frac{dn^+}{dt} = \frac{dn^-}{dt} = -\alpha n^+ n^- \quad [\alpha] = 10^{-6} - 10^{-8}\frac{cm^3}{s}$
- Recombination is the dominant process if no or very small voltage is applied

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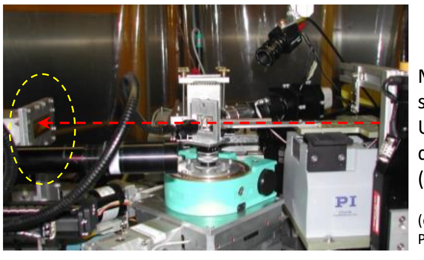
Ionization chamber

- V ≈ 1 mm^3 – 0,1 m^3
- Current or pulse mode of operation
- I ≈ 10^{-12} A or smaller   R ≈ 10^{16} Ω
- Extremely small leakage current < 1% (!) ⇨ special insulation required
- Used in many applications, such as dosimetry, reactors, neutron detection, calibrators, smoke detectors
- Easily accommodates converter material for neutron conversion
- Processing of signal is real challenge as piko-amper measurement is needed

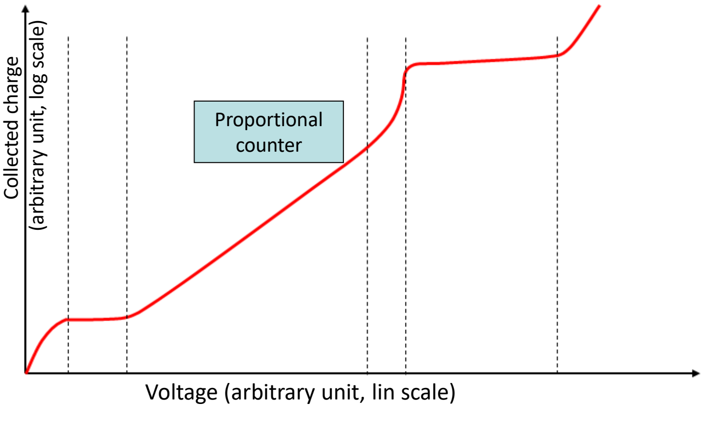


Measurement of sincrotron X-ray source Using a gas-ionization detector (Hamburg DESY)

(courtesy of Prof. Imre Szalóki)

Proportional counter:

larger pulses, less requirement on amplification



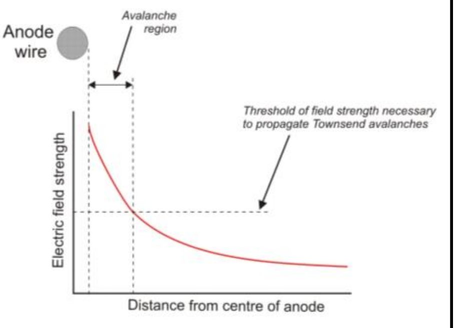
Proportional counter

- Electric field inside a gas-filled detector tube (cylindrical shape) depends on distance from anode wire:

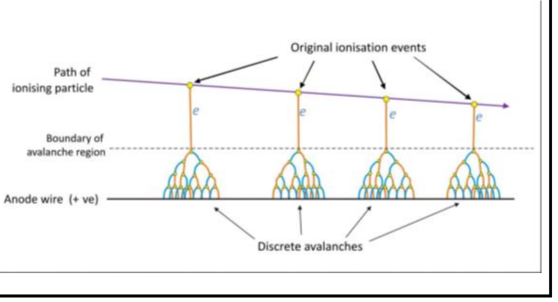
$$E(r) = \frac{U_0}{r\ln(r_k/r_a)} \qquad \frac{r_k}{r_a} \approx 100 - 300$$

where r_k = radius of anode wire, r_a = radius of cathode cylinder

- U = 2000V, r_a = 80 µm, r_k = 10 mm ⇨ $E \approx 10^6$ V/m
- Significant gas-multiplication ($10^3 - 10^5$) in the vicinity of anode wire, where electric field is highest is wire is thin
- Limitation on the radius of anode wire ⇨ to limit multiplication
- Energy resolution around 1%
- Applications: position-sensitive measurement in large particle detectors, surface contamination, reactors



https://en.wikipedia.org/wiki/Proportional_counter



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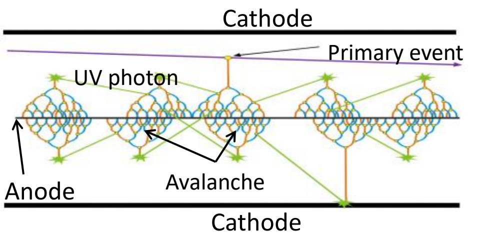
Geiger-Müller counter (GM tube)

- Increasing voltage ⇨ very large charge multiplication ≈ $10^6 - 10^7$
- Quenching is necessary due to the large number of ions
- Energy-independent detection ⇨ G-M tube is not a spectrometer ⇨ GM tube is not linear at all(!)
- High output signal (≈ 1-2 V) ⇨ simple electronics, even a simple speaker is sufficient to "hear the counts"
- Robust, hard-to-damage devices with lots of applications; 50 year old and older devices still operate
- Dosimeters installed at BME Training Reactor are GM-tube based devices



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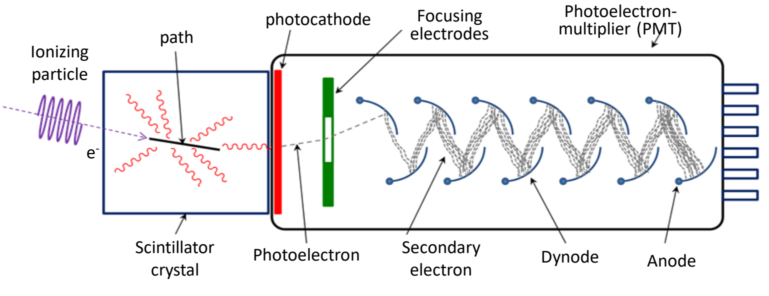
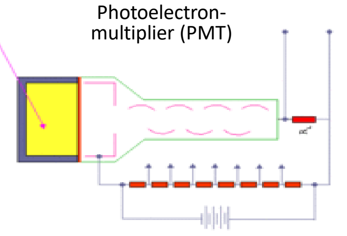
Specific detector types:

Scintillation detectors

Operating principle of scintillation detectors

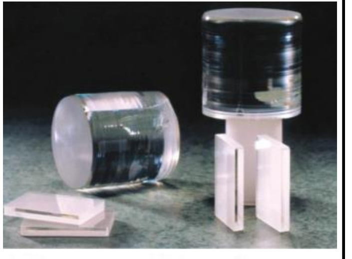




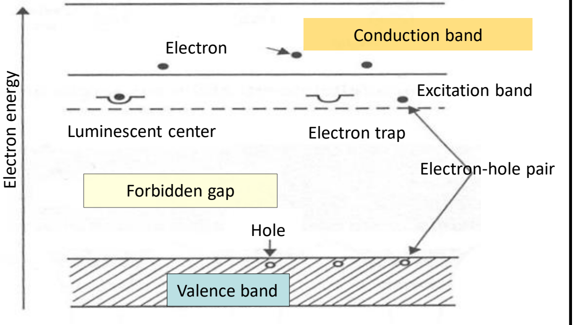
- Interaction of *scintillator material* + ionizing radiation produces electrons ⇨ electrons excite scintillator
- Relaxation of excited states ⇨ create visible light emissions
- Light photons reach the photocathode of the PMT tube ⇨ electrons are created (few) (by photoelectric effect)

- Electrons are accelerated by gradually increasing voltage of dynodes (electric field) ⇨ upon reaching dynodes, they create more electrons, thereby eventually producing a substantial voltage pulse

The mechanism of scintillation



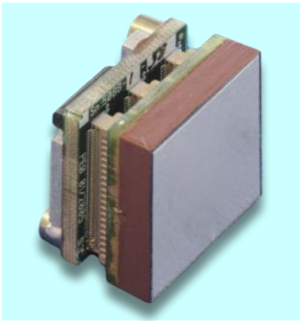
- The *scintillation process* (in crystals) is due to the electronic band structure.
- An incoming particle excites an electron from the valence band to either the conduction band or the exciton band.
- This leaves a hole behind in the valence band.
- Impurities are normally materials added to the "base" material in small fractions, such as $10^{-3} – 10^{-4}$ to create electronic levels (traps) in the forbidden gap.
- The excitons are loosely bound electron-hole paires which wander through the crystal lattice until they are captured as a whole by impurity centers.
- The latter then rapidly de-excite by emitting scintillation light (fast component). The activator impurities are typically chosen such that emitted light is in the visible range (where photomultiplier photocathodes are effective). There is also a delayed light component.

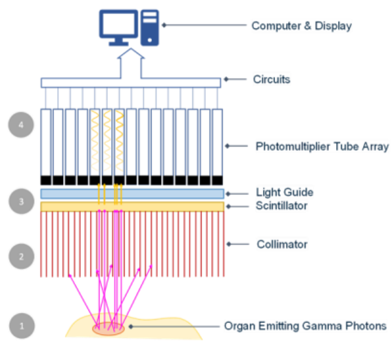


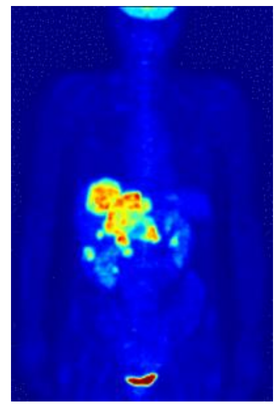
Sodium iodide: NaI(Tl) as the most common scintillator material

- Linear
- Fast
- Easy to produce, easy to handle and easy to work with
- Mainly for gamma radiation detection
- Widely used In nuclear medicine devices

Gamma-camera: a medial application







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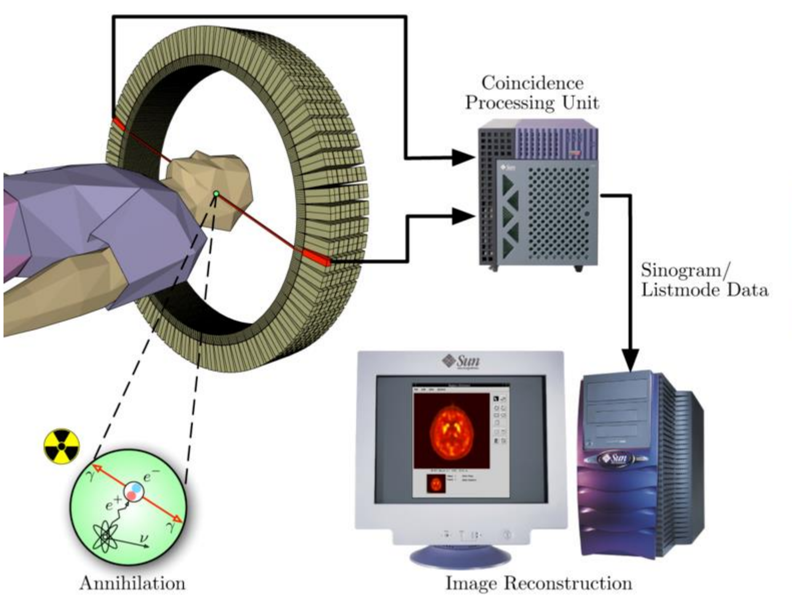
PET= positron emission tomography



- Coincidence counting
- A positron emitter isotope in administered to the body of the patient
- Very large number of scintillation detectors detect gamma radiation produced by the annihilation of positrons
- Spatial distribution of isotope (molecule carrying the isotope) can be determined
- Can be used to detect brain activity, inflammations, tumours

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Nuclear electronics
or
How to process signals from detectors

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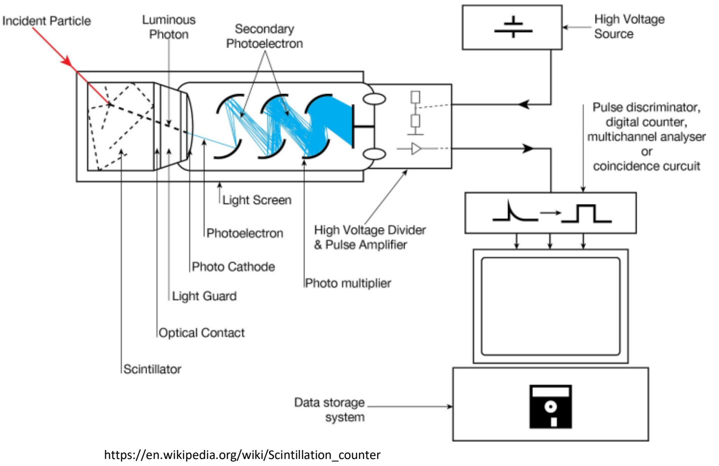
Nuclear electronics

- Electronic circuits are needed to process the signal from detectors
- Strick requirements for the electronics make them expensive



https://en.wikipedia.org/wiki/Scintillation_counter

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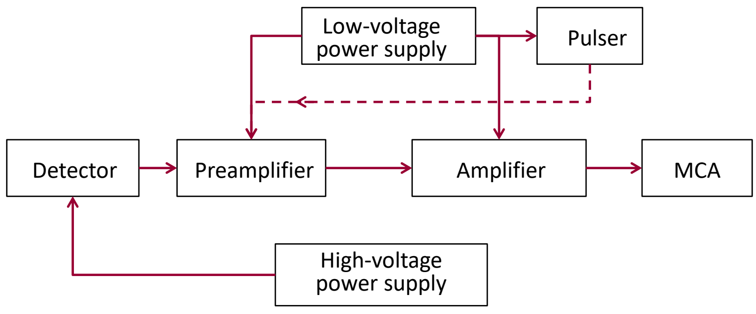
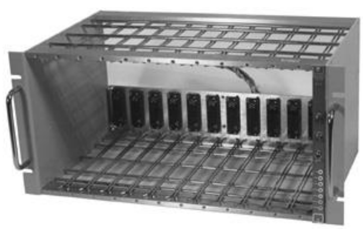
Nuclear electronics



Main constituents of signal processing system

- High- and low-pass filters (pulse shaping)
- Baseline shift restorer
- Pile-up rejection circuit
- Voltage or charge-sensitive preamplifier
- Amplifier
- Analog-to-digital converter
- Multichannel analyzer or digital spectrum analyzer
- Power supply

- NIM = Nuclear Instrument Module
- Power supply: 6,12,24 V
- Modular design
- Standard units: amplifier, pulser, high-voltage power supply,…



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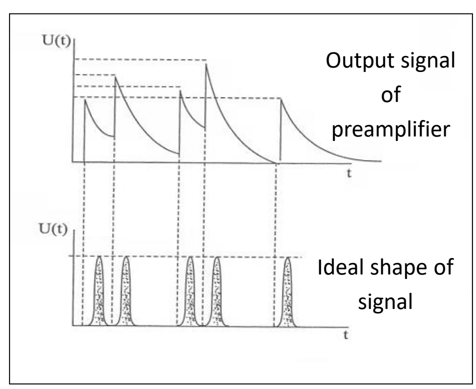
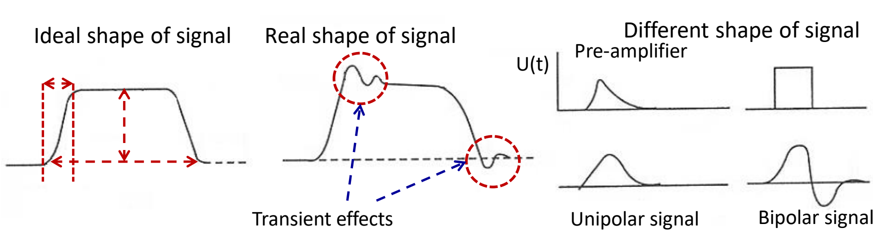
Processing of electronic signals: shaping

- Signal/noise optimization
- Charge collecting ≈ 50 µs
- Signal conditioning:
  application of RC, CR filters
- Time-constant $\tau = RC$
- Cable impedance





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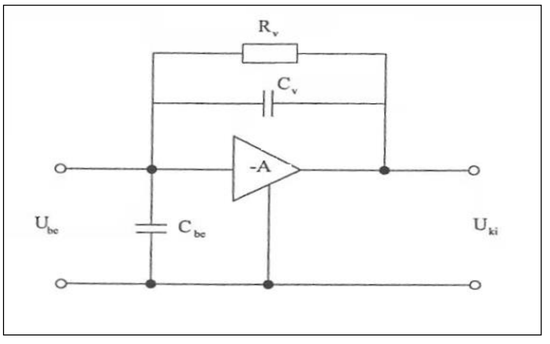
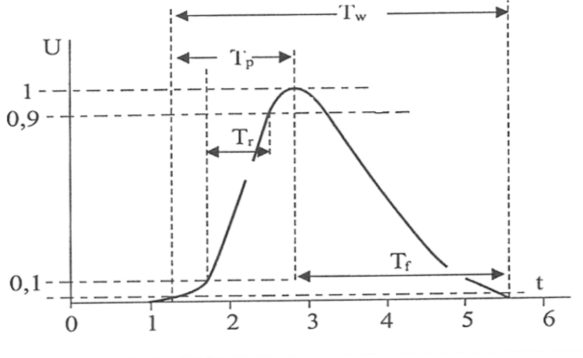
Pre-amplifier



- The pre-amplifier is set close to the detector
- Optical feedback to eliminate the source of thermal noise
- Pre-amplifiers generally cooled cryogenically
- TEST input ⇨ pulser ⇨ electronic noise analysis

Linear amplifier

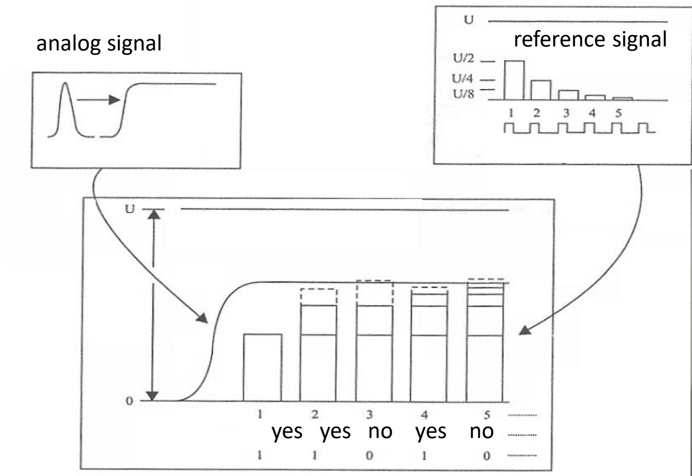
- Gain: 10 - 5000
- Signal/noise improvement
- Pulse shaping: improving energy resolution
- Shape: unipolar (better signal/noise), bipolar
- Polarity: positive, negative, U < 10V



- Tr rise time
- Tp peaking time
- Tw pulse length
- Tf decay time
- Shaping time 0.5-12 µs
  - Scintillation 0.5-1 µs
  - Si(Li) 10 µs
  - Ge 3-6 µs

Analog-to-digital converter: Successive Approximation

- Constant short dead time 1-5 µs
- Integral nonlinearity <1%



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